

Performance of GPU-Accelerated VASP 6.5.1 on a Dual NVIDIA H100 Node:

Materials Project PBE and r²SCAN Workflows, Finite-Displacement Phonons, and Concurrent Runs on One GPU

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Abstract

We benchmark the GPU (OpenACC) build of VASP 6.5.1 on a shared node with two NVIDIA H100 NVL GPUs. Single-point and full cell relaxation calculations are run on supercell series of Si, Al, and MgO from 1 to 64 atoms, using the Materials Project PBE and r²SCAN input sets. We report wall time, time per self-consistent step, GPU memory, the cost of r²SCAN relative to PBE, the speedup from one to two GPUs with two parallelization strategies, run-to-run reproducibility, and agreement with published Materials Project energies. For these sizes, one calculation leaves the GPUs largely idle: mean utilization stays between 11 and 51 %, and peak memory between 2.8 and 12.2 GB out of 94 GB. Time per SCF step grows sublinearly with the number of atoms because the k -point count decreases in larger supercells. r²SCAN costs 0.9 to 7.9 $\times$ more than PBE for a single point and 1.5 to 15.9 $\times$ more for a relaxation. Two GPUs give a speedup of 1.0 to 1.9 $\times$; distributing k -points (KPAR=2) is the better choice up to about 32 atoms, and distributing bands (KPAR=1) can overtake it for 64-atom cells with 3 or 4 k -points. Timings are reproducible to within 1 %. Energies agree with Materials Project to within a few meV/atom, except for two understood cases: an older Mg PAW potential in the Materials Project PBE data, and coarse k -point sampling of metallic Al. Complete finite-displacement phonon workflows of Si, Al, and MgO take 0.5 to 1.7 h on one GPU and run 1.4 to 1.6 $\times$ faster on two. Finally, running several small calculations at once on one GPU recovers much of the idle capacity, but only with the NVIDIA Multi-Process Service (MPS): eight concurrent single points raise throughput by 4.1 to 5.8 $\times$ with MPS, whereas without MPS throughput stays between 0.55 and 1.35 $\times$.

1 Introduction

Plane-wave density-functional theory (DFT) codes such as VASP [1] now run on GPUs, and high-throughput databases such as Materials Project [2] rely on standardized workflows run on thousands of structures, most of them with fewer than 100 atoms. A small shared node with two recent data-center GPUs raises practical questions for its users: which system sizes make good use of the hardware, what the standard workflows cost, and how best to use one or two GPUs. This report

answers these questions for the GPU build of VASP 6.5.1 on such a node, using the Materials Project PBE and r²SCAN input sets unchanged.

This study addresses the following questions:

1. How does computing time scale with the number of atoms, from 1 to 64?
2. How much more does r²SCAN cost than PBE, per self-consistent (SCF) step and in total?
3. How much does a full relaxation cost compared with a single point?
4. What is gained by going from one to two GPUs, and is it better to distribute k -points (KPAR=2) or bands and plane waves (KPAR=1)?
5. How much GPU memory is needed as a function of system size?
6. Are timings reproducible from one job to the next?
7. Are the computed energies consistent with those published by Materials Project?
8. What is the time-to-solution of a complete finite-displacement phonon calculation on one and two GPUs?
9. Can several small calculations share one GPU to recover its idle capacity, with and without the NVIDIA Multi-Process Service (MPS)?

2 Hardware and Software

Table 1: Test system.

Component	Specification
GPUs	2 × NVIDIA H100 NVL, 94 GB each (driver 595.84)
CPU	AMD EPYC 7662, 32 cores assigned to the virtual machine
Host memory	62 GB
Per GPU (Slurm)	16 cores, ~28 GB host memory, no time limit
DFT code	VASP 6.5.1 (release of March 10, 2025), OpenACC GPU build compiled on September 23, 2026 (<code>vasp_std</code> , <code>vasp_gam</code>)
Toolchain	NVIDIA HPC SDK 26.3, HPC-X (CUDA-aware Open MPI), Intel oneMKL
Execution	one MPI rank per GPU, <code>OMP_NUM_THREADS=1</code>
Workflow tools	pymatgen 2026.9.23 [3], phonopy 4.6.0 [4, 5]

The node is shared with other users through Slurm. To leave the second GPU available, study jobs were submitted as job arrays running one calculation at a time. One-GPU and two-GPU series were never submitted concurrently. For each calculation, the jobs running on the node at start time were recorded, since they may affect timings (Section 3.4).

3 Methods

3.1 Structures

All structures were retrieved from Materials Project [2] (Table 2). To isolate the effect of system size at constant chemistry, each material is studied as a series of supercells of increasing size, 14 structures in total. The MP structures are already relaxed with PBE.

Table 2: Scaling series.

Material	MP ID	Type	Sizes (atoms)
Si, diamond	mp-149	semiconductor (PBE gap 0.61 eV)	2 (primitive), 8, 16, 32, 64
Al, fcc	mp-134	metal	1 (primitive), 4, 32, 64
MgO, rock salt	mp-1265	insulator (PBE gap 4.43 eV)	2 (primitive), 8, 16, 32, 64

3.2 Electronic-structure settings

All calculations use the projector augmented-wave (PAW) method [6, 7] as implemented in VASP [1]. Inputs were generated with the Materials Project input sets implemented in pymatgen [3], without changing any physical parameter (Table 3). The PBE sets [8] apply the MP Hubbard U values [9] to oxides and fluorides of Co, Cr, Fe, Mn, Mo, Ni, and V (none of the present materials is affected). The r^2 SCAN sets [10] follow the MP meta-GGA workflow [11], in which the k -point spacing is chosen from the MP band gap. All calculations are spin-polarized (ISPIN=2), with PREC=Accurate and LASPH=TRUE. The complete INCAR files are given in Appendix A.

Table 3: Calculation series and main parameters. Brillouin-zone sampling is expressed per reciprocal-lattice volume (PBE) or as KSPACING (r^2 SCAN).

Series	pymatgen set	Main parameters
pbe_sp	MPStaticSet	PBE; ENCUT=520 eV; 100 k -points per $\text{\AA}^{-3}$; EDIFF= 5×10^{-5} eV per atom; ALGO=Fast; LREAL=FALSE.
pbe_relax	MPRelaxSet	as above, with ISIF=3, IBRION=2; 64 k -points per $\text{\AA}^{-3}$; LREAL=Auto
r2scan_sp	MPScanStaticSet	METAGGA=R2SCAN; ENCUT=680 eV; KSPACING from 0.22 to $0.44 \text{\AA}^{-1}$ depending on the MP gap; EDIFF= 1×10^{-5} eV
r2scan_relax	MPScanRelaxSet	as above, with ISIF=3, ALGO=All, EDIFFG= $-0.02 \text{ eV/\AA}$

The following deviations from the MP methodology apply identically to every calculation:

- **Pseudopotentials.** The PBE_54 PAW set installed on the machine is used for both functionals. MP uses an older PBE set for its GGA calculations, which is identical for almost all elements; W_pv, absent from PBE_54, would be replaced by W_sv. Among the present materials, only Mg differs: MP used Mg_pv_06Sep2000, while PBE_54 provides Mg_pv_13Apr2007 (Section 4.7).
- **Output.** Writing of WAVECAR, CHGCAR, AECCAR, LOCPOT, and ELFCAR is disabled, so that the benchmark measures computation rather than I/O.
- **Parallelization tags.** NCORE is removed, since the GPU build forces it to 1; KPAR is set by the parallelization variant (Section 3.3).
- **Small k -point meshes.** When a mesh has fewer than 4 irreducible k -points, the tetrahedron method (ISMEAR=-5) is replaced by Gaussian smearing (ISMEAR=0, SIGMA=0.05 eV), which is also the correction applied by custodian in MP workflows.

In the supercells, the k -point density per atom is kept constant, as in the MP methodology, so the number of k -points decreases as the cell grows. Relaxations start from the MP structures,

which are already relaxed with PBE: PBE relaxations therefore converge in a few ionic steps, and r²SCAN relaxations need a few more.

3.3 Parallelization variants

Three variants are compared (Table 4), with identical physical inputs. With two GPUs, KPAR=1 distributes bands and plane-wave coefficients across both devices, while KPAR=2 assigns one group of k -points to each GPU. The g2k2 variant is skipped when a structure has fewer than two irreducible k -points.

Table 4: Parallelization variants.

Variant	GPUs	MPI ranks	KPAR
g1	1	1	1
g2k1	2	2	1
g2k2	2	2	2

3.4 Measurement protocol

Each calculation runs as one Slurm job-array task launched with `mpirun -np  $N_{\text{GPU}}$  -bind-to none vasp_std`. The following quantities are recorded:

- **Problem size:** number of atoms, NKPTS, NBANDS, NPLWV, NELECT, ISPIN.
- **Time:** total time (“Elapsed time” in OUTCAR, and job wall time); median time per SCF step (LOOP lines); mean time per ionic step (LOOP+ lines); numbers of SCF and ionic steps.
- **Memory and utilization:** peak GPU memory and mean GPU utilization, sampled every 2s with `nvidia-smi`; peak host memory.
- **Energy:** energy per atom and its difference from the uncorrected MP energy of the same functional, in meV/atom.
- **Context:** other jobs running on the node at start time.

The speedup of a two-GPU variant is the ratio of the one-GPU total time to the two-GPU total time. Reproducibility was assessed by repeating six single-point calculations (Si 8 atoms, Al 32 atoms, and MgO 32 atoms, with PBE and r²SCAN) three times each on one GPU.

The study was carried out in phases, each submitted separately from the smallest to the largest system (Table 5).

3.5 Finite-displacement phonons

Harmonic phonons of the primitive cells of Si, Al, and MgO are computed with the finite-displacement method using phonopy [4, 5]. Each material is run as one automated workflow per Slurm task, with identical inputs on one GPU (g1) and two GPUs (g2k1, KPAR=1), so that only the hardware changes.

Table 5: Study phases.

Phase	Content	GPUs	Calculations
0	Workflow validation on 1–2 atoms	1	5
1	Reproducibility, 3 repeats	1	18
2	Single points, PBE and r ² SCAN	1	28
3	Relaxations, PBE and r ² SCAN	1	28
4	All four series, g2k1 and g2k2	2	112
5	Finite-displacement phonons, g1 and g2k1	1, 2	6 workflows
6	Concurrent single points on one GPU, with and without MPS	1	26 batches, 100

Settings. PBE with PBE_54 PAW potentials, ENCUT=520 eV, PREC=Accurate, EDIFF= 1×10^{-8} eV, ADDGRID=.TRUE., LREAL=.FALSE., LASPH=.TRUE., ALGO=Normal, non-spin-polarized. Gaussian smearing with $\sigma = 0.01$ eV is used for Si and MgO, and first-order Methfessel–Paxton smearing with $\sigma = 0.2$ eV for Al. Γ -centered k -meshes are built with $N_i = \max(1, \text{round}(L |\mathbf{b}_i|))$, where $\mathbf{b}_i$ are the reciprocal lattice vectors (without the 2π factor) and $L = 50$ Å for insulators and 80 Å for the metal.

Workflow.

1. **Tight relaxation** of the primitive cell (ISIF=3, EDIFFG= -1×10^{-3} eV/Å). The relaxation is restarted from the previous CONTCAR, which resets the plane-wave basis, until a pass completes in at most two ionic steps (at most four passes).
2. **Born effective charges and ε_∞** for polar materials (MgO) from density-functional perturbation theory (LEPSILON), with a finite-field fallback (LCALCEPS). They are used for the non-analytical term correction.
3. **Force calculations** for the symmetry-irreducible displacements (0.01 Å) in two diagonal supercell sizes, to check convergence (Table 6).
4. **Post-processing:** symmetrized force constants, band structure along the seekpath path [12], density of states on a 30^3 q -mesh, and thermodynamic properties from 0 to 1,000 K.

Table 6: Phonon supercells.

Material	Relax. k -mesh	Supercells (atoms)	Displacements	Supercell k -meshes
Si	16^3	3^3 (54), 4^3 (128)	1 per cell	5^3 , 4^3
Al	34^3	4^3 (64), 5^3 (125)	1 per cell	9^3 , 7^3
MgO	21^3	3^3 (54), 4^3 (128)	2 per cell	7^3 , 5^3

Metrics. VASP wall time of every step and GPU-hours; time-to-solution from submission to result, queue time included; quality checks: acoustic frequencies at Γ , lowest frequency on the q -mesh (imaginary modes), and residual force drift.

3.6 Concurrent runs on one GPU

Four single points of the one-GPU campaign were rerun as batches of N identical copies sharing one GPU: PBE Si (8 atoms) and MgO (32 atoms), and r²SCAN Al (32 atoms) with $N = 2, 4$, and 8; r²SCAN Si (64 atoms) with $N = 2$ and 4, limited by the ~ 28 GB of host memory available per GPU. The inputs are those of the **g1** runs, unchanged. Each batch runs in a single Slurm job with one GPU and 8 cores, since Slurm does not share a GPU between jobs on this node; each copy is one MPI rank with `OMP_NUM_THREADS=1`. Two modes are compared. Without MPS, the CUDA contexts of the N processes share the GPU by time slicing, the default behavior. With MPS, a daemon private to the job lets kernels from the N processes run on the GPU at the same time. A run of one copy under MPS checks that the daemon itself costs nothing.

The throughput gain of a batch is $N t_{\text{ref}}/T_{\text{batch}}$, where t_{ref} is the mean one-GPU time of the calculation run alone (Section 4.1) and T_{batch} is the time from the first start to the last end of the batch; the ideal value is N . The slowdown of each calculation is its time in the batch divided by t_{ref} .

4 Results

4.1 Scaling with system size on one GPU

All 56 one-GPU calculations completed, and all 28 relaxations converged. Table 7 gives the median time per SCF step and the total time; Figs. 1 and 2 plot them against the number of atoms.

Table 7: One-GPU timings (**g1**): median time per SCF step t_{SCF} (s) and total time t_{tot} (s, OUTCAR elapsed time). Phase-1 cases are averaged over three repeats.

	Atoms	PBE SP		PBE relax		r ² SCAN SP		r ² SCAN relax	
		t_{SCF}	t_{tot}	t_{SCF}	t_{tot}	t_{SCF}	t_{tot}	t_{SCF}	t_{tot}
Si	2	1.05	15	0.74	20	1.75	24	2.60	31
Si	8	1.27	18	1.20	31	3.26	42	5.32	92
Si	16	2.75	40	2.72	69	7.35	101	11.79	215
Si	32	4.75	69	4.57	118	13.34	184	21.97	397
Si	64	7.46	110	6.68	175	23.59	324	38.78	693
Al	1	0.99	14	0.68	14	3.37	39	4.66	101
Al	4	0.96	13	0.89	12	3.75	49	5.29	166
Al	32	2.78	34	2.68	57	12.66	181	21.37	594
Al	64	6.17	83	5.50	122	33.72	654	51.15	1943
MgO	2	2.53	36	1.69	46	1.97	31	3.07	66
MgO	8	3.44	51	2.06	59	4.88	76	7.89	270
MgO	16	6.89	97	4.63	127	11.13	173	17.40	696
MgO	32	10.06	138	7.89	218	19.56	337	31.55	1528
MgO	64	12.32	163	11.13	309	29.44	470	50.40	2864

Scaling. Times are short: every PBE calculation finishes in less than 5.5 min, and the longest calculation, the r²SCAN relaxation of 64-atom MgO, takes 48 min. Fitting $t_{\text{SCF}} \propto N^\alpha$ between 8 and 64 atoms for Si and MgO gives $\alpha = 0.6\text{--}0.85$ with PBE and $0.85\text{--}0.95$ with r²SCAN: the time per step grows more slowly than the number of atoms. This is mostly a consequence of the MP k -point density, which is fixed per reciprocal volume, so that the number of irreducible k -points

Temps par pas SCF (s, médiane) sur 1 GPU H100

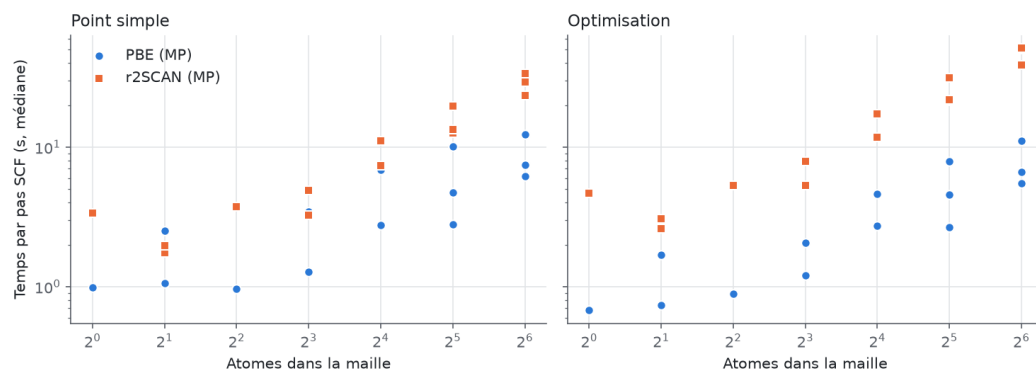


Figure 1: Median time per SCF step versus number of atoms on one GPU.

Temps total (s) sur 1 GPU H100

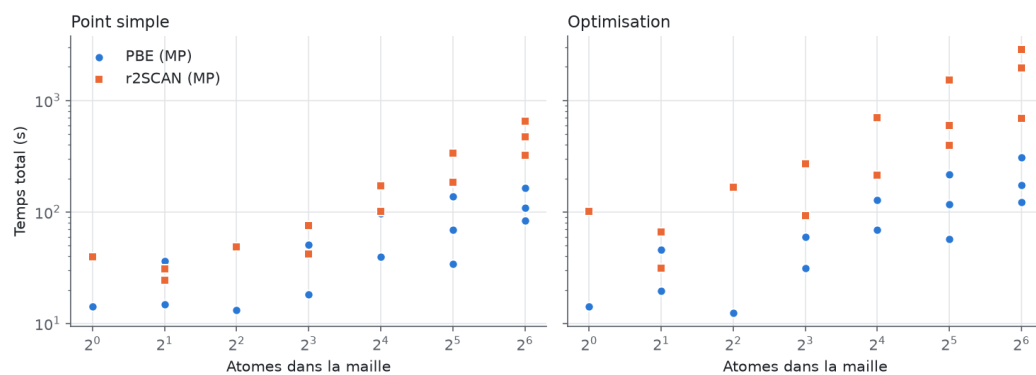


Figure 2: Total time versus number of atoms on one GPU.

drops from 10–20 at 8 atoms to 4 at 64 atoms. Per k -point, the exponent is 1.25–1.4, still far below the N^2 – N^3 behavior expected once the GPU is saturated. For the smallest cells, times are almost flat (about 1 s per PBE SCF step for Si and Al up to 8 atoms): fixed overheads dominate.

4.2 Cost of r²SCAN relative to PBE

Table 8 gives the ratio of r²SCAN to PBE times. It measures the cost of the MP r²SCAN workflow relative to the MP PBE workflow as used in practice, not of the functional alone: r²SCAN also uses a higher cutoff (680 instead of 520 eV) and, for Al, a denser k -mesh (12 instead of 3 k -points at 64 atoms).

Table 8: Cost of r²SCAN relative to PBE on one GPU: ratio of total times and of times per SCF step.

	Atoms	Single point		Relaxation	
		Total	Per SCF step	Total	Per SCF step
Si	2	1.6	1.7	1.6	3.5
Si	8	2.3	2.6	2.9	4.4
Si	16	2.6	2.7	3.1	4.3
Si	32	2.6	2.8	3.4	4.8
Si	64	3.0	3.2	4.0	5.8
Al	1	2.8	3.4	7.1	6.9
Al	4	3.7	3.9	13.3	5.9
Al	32	5.3	4.6	10.4	8.0
Al	64	7.9	5.5	15.9	9.3
MgO	2	0.9	0.8	1.5	1.8
MgO	8	1.5	1.4	4.5	3.8
MgO	16	1.8	1.6	5.5	3.8
MgO	32	2.4	1.9	7.0	4.0
MgO	64	2.9	2.4	9.3	4.5

The ratio grows with system size. Per SCF step in single points, r²SCAN costs 1.7 to 3.2× more for Si, 3.4 to 5.5× for Al (with a denser k -mesh), and 0.8 to 2.4× for MgO; for the MgO primitive cell, the r²SCAN step is cheaper because its k -mesh has 16 instead of 56 irreducible points. For relaxations the total ratio is larger (up to 15.9× for Al and 9.3× for MgO), because r²SCAN relaxations need more SCF steps (Section 4.3).

4.3 Relaxation versus single point

Table 9 compares relaxations with single points. Since the MP structures are already relaxed with PBE, PBE relaxations take 1 to 3 ionic steps and cost 0.9 to 1.9× a single point; the PBE relaxation set also uses a coarser k -mesh and LREAL=Auto, which makes each SCF step cheaper. r²SCAN relaxations take 1 to 4 ionic steps, since the cell adjusts to the new functional, and 13 to 46 SCF steps in total, with ALGO=All; they cost 1.3 to 6.1× a single point, the ratio being largest for MgO.

4.4 GPU memory

Table 10 gives the peak GPU memory. About 2.9 GB is used even for one atom (CUDA context, libraries, and fixed VASP work arrays). Above this baseline, memory grows with the number of

Table 9: Relaxations versus single points on one GPU: numbers of ionic and SCF steps of the relaxation, and ratio of total times (relaxation / single point).

	Atoms	PBE			r ² SCAN		
		Ionic	SCF	Ratio	Ionic	SCF	Ratio
Si	2	3	19	1.31	1	13	1.28
Si	8	3	20	1.73	2	18	2.19
Si	16	3	20	1.75	2	19	2.12
Si	32	3	20	1.70	2	19	2.16
Si	64	3	20	1.60	2	19	2.14
Al	1	3	13	1.00	3	20	2.57
Al	4	1	10	0.94	4	27	3.39
Al	32	3	14	1.68	3	26	3.28
Al	64	3	16	1.47	3	31	2.97
MgO	2	3	23	1.26	2	21	2.15
MgO	8	3	25	1.17	3	31	3.56
MgO	16	3	24	1.31	3	34	4.03
MgO	32	3	24	1.58	3	39	4.54
MgO	64	3	24	1.89	3	46	6.10

plane waves, bands, and k -points: with PBE it never exceeds 5.2 GB, and with r²SCAN it reaches 9.2 GB for the 64-atom Si relaxation and 12.2 GB for the 64-atom Al relaxation (12 k -points). All values are far below the 94 GB of each GPU. Peak host memory stays below 9.5 GB, within the $\sim$ 28 GB allocated per GPU.

Table 10: Peak GPU memory on one GPU (GB), sampled every 2s with `nvidia-smi`.

	Atoms	PBE SP	PBE relax	r ² SCAN SP	r ² SCAN relax
Si	2	2.9	2.9	3.0	3.1
Si	8	3.0	3.0	3.2	3.3
Si	16	3.2	3.2	3.6	4.1
Si	32	3.8	3.8	4.6	5.9
Si	64	5.2	5.1	7.0	9.2
Al	1	2.8	2.8	2.9	3.0
Al	4	2.9	2.9	3.0	3.1
Al	32	3.3	3.3	4.1	5.1
Al	64	4.3	4.3	7.6	12.2
MgO	2	2.9	2.9	2.9	3.0
MgO	8	2.9	2.9	3.0	3.1
MgO	16	3.2	3.1	3.2	3.6
MgO	32	3.6	3.5	3.9	5.0
MgO	64	4.3	4.3	4.8	6.5

4.5 Reproducibility

The six single points of phase 1 were each run three times on one GPU (Table 11). The relative spread of the total time (standard deviation over mean) is at most 1.0%, so differences of more than a few percent between variants are significant.

Table 11: Reproducibility of total times over three repeats on one GPU.

Series	Material	Atoms	Mean time (s)	Relative spread
PBE SP	Si	8	18.1	1.0 %
PBE SP	Al	32	34.0	0.6 %
PBE SP	MgO	32	138.1	0.2 %
r ² SCAN SP	Si	8	42.1	0.3 %
r ² SCAN SP	Al	32	180.7	0.2 %
r ² SCAN SP	MgO	32	337.0	0.5 %

4.6 Two GPUs: speedup and KPAR

All 112 two-GPU calculations completed, and all 56 two-GPU relaxations converged. Total energies agree with the one-GPU values to within 0.01 meV/atom. Table 12 and Fig. 3 give the speedup over one GPU.

Table 12: Speedup on two GPUs (one-GPU total time divided by two-GPU total time). Bold: variant faster by more than 2 %. Run-to-run spread is at most 1 % (Section 4.5).

	Atoms	PBE SP		PBE relax		r ² SCAN SP		r ² SCAN relax	
		g2k1	g2k2	g2k1	g2k2	g2k1	g2k2	g2k1	g2k2
Si	2	1.12	1.49	1.16	1.57	1.26	1.57	1.37	1.57
Si	8	1.31	1.52	1.38	1.62	1.52	1.64	1.52	1.69
Si	16	1.45	1.66	1.50	1.73	1.64	1.72	1.62	1.72
Si	32	1.51	1.59	1.54	1.61	1.63	1.61	1.61	1.59
Si	64	1.66	1.60	1.66	1.60	1.66	1.57	1.57	1.54
Al	1	1.03	1.50	0.99	1.42	1.24	1.72	1.19	1.75
Al	4	1.15	1.48	1.08	1.42	1.45	1.71	1.57	1.77
Al	32	1.49	1.56	1.58	1.59	1.62	1.68	1.62	1.65
Al	64	1.67	1.47	1.68	1.50	1.69	1.69	1.57	1.55
MgO	2	1.30	1.70	1.31	1.68	1.42	1.58	1.42	1.60
MgO	8	1.45	1.79	1.54	1.80	1.66	1.80	1.46	1.85
MgO	16	1.54	1.84	1.60	1.85	1.75	1.87	1.77	1.89
MgO	32	1.59	1.84	1.65	1.77	1.79	1.78	1.79	1.80
MgO	64	1.69	1.80	1.73	1.82	1.77	1.81	1.63	1.79

Magnitude. The speedup ranges from 0.99 to 1.79 $\times$ with KPAR=1 and from 1.42 to 1.89 $\times$ with KPAR=2. Two GPUs never halve the run time.

Choice of KPAR. Up to 16 atoms, the cells have 10 to 84 irreducible k -points and few bands: distributing k -points (KPAR=2) is clearly faster than distributing bands and plane waves (KPAR=1), which brings almost nothing for the 1-atom Al cell. At 32 atoms the gap narrows: KPAR=2 stays ahead in most PBE cases, and both variants agree within 3 % with r²SCAN. At 64 atoms, KPAR=1 becomes faster for Si (4 k -points) and for Al with PBE (3 k -points, split 2 + 1 between the GPUs with KPAR=2), while MgO (4 k -points) remains faster with KPAR=2. Over the 56 cases, KPAR=2 is faster in 42, KPAR=1 in 6 (all with 64 atoms), and the two agree within 2 % in 8.

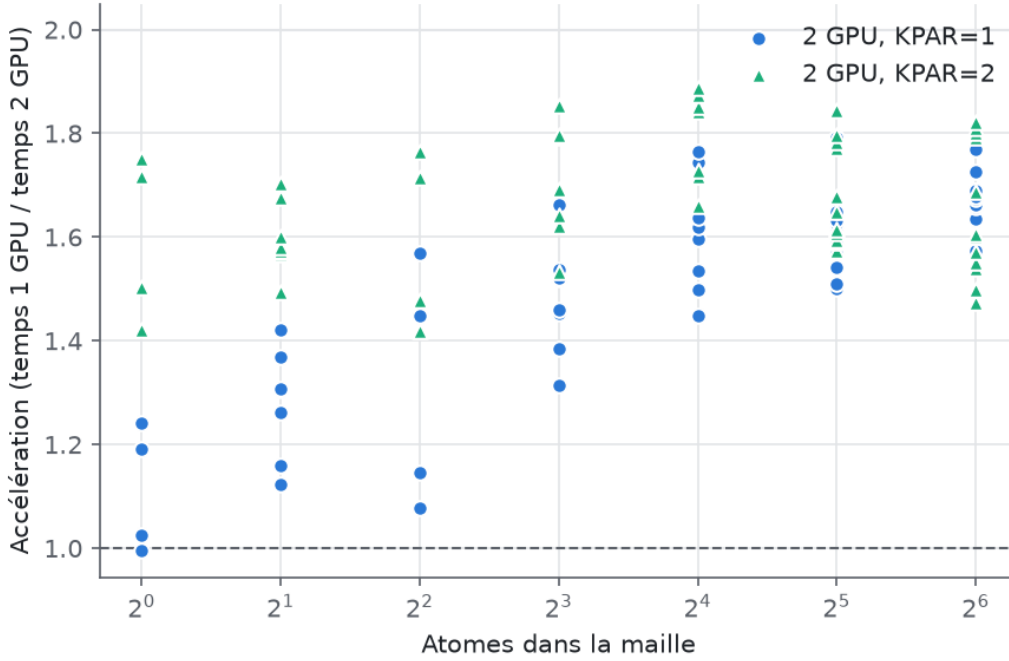


Figure 3: Speedup on two GPUs versus number of atoms, for KPAR=1 (g2k1) and KPAR=2 (g2k2).

4.7 Agreement with Materials Project energies

Table 13 gives the difference between the computed energy per atom and the uncorrected MP energy of the same functional. Energies computed on two GPUs agree with the one-GPU values to within 0.01 meV/atom.

Si and r²SCAN. Si agrees with MP to within 5 meV/atom with both functionals, and all three materials agree to within 7 meV/atom with r²SCAN.

MgO with PBE. The offset is constant, +60 meV/atom for single points and +50 for relaxations, independent of cell size. It comes from the Mg pseudopotential: according to the MP task data, the MP GGA calculation of mp-1265 used PAW_PBE Mg_pv_06Sep2000, whereas the PBE_54 set used here provides Mg_pv_13Apr2007 (the O potential is the same). The MP r²SCAN calculation used Mg_pv_13Apr2007, as here, and the r²SCAN energies agree to within 2 meV/atom. The difference between single points and relaxations comes from the different k -meshes and LREAL settings of the two sets.

Al with PBE. The deviation varies with cell size: from -0.5 to -22 meV/atom for single points, largest for 32 atoms, and from $+18$ to -22 meV/atom for relaxations. Al is a metal, whose total energy converges slowly with k -point sampling. In the supercells, the MP k -point density per atom gives coarse meshes ($3 \times 3 \times 3$, 4 irreducible points, for 32 atoms), where the tetrahedron method is still used; for 64 atoms, the mesh has 3 points and Gaussian smearing is used instead (Section 3.2). The relaxation set, with its lower k -point density, shows larger deviations. These values reflect the sampling of the MP input sets in supercells, not a problem with the GPU build.

Table 13: Energy difference from the uncorrected MP energy of the same functional (meV/atom), one GPU.

	Atoms	PBE SP	PBE relax	r ² SCAN SP	r ² SCAN relax
Si	2	3.4	4.6	2.8	2.4
Si	8	1.9	2.5	-0.9	-1.5
Si	16	1.8	2.4	-1.1	-1.7
Si	32	2.4	2.4	-1.2	-1.8
Si	64	4.0	2.5	-1.2	-1.8
Al	1	-0.5	18.1	1.5	-3.0
Al	4	-2.2	-19.1	-1.7	-6.5
Al	32	-22.1	-22.3	-0.1	-6.1
Al	64	-3.8	-5.2	-1.5	-6.1
MgO	2	59.9	50.4	0.9	0.5
MgO	8	59.9	50.4	1.8	1.2
MgO	16	59.9	50.4	1.8	1.2
MgO	32	59.9	50.4	1.8	1.1
MgO	64	59.9	50.4	1.8	1.1

4.8 Phonons

Figure 4 shows the PBE phonon dispersions computed on one GPU (**g1**) for both supercell sizes. No imaginary frequency appears: on the 30^3 q -mesh, the lowest frequency away from Γ is positive for all three materials, and the acoustic branches go to zero at Γ . The highest optical frequency of Si at Γ is 503 cm^{-1} ; the Al spectrum extends up to 322 cm^{-1} . For MgO, the non-analytical term correction (Born charge $Z^* = 1.99$, $\varepsilon_\infty = 3.24$) splits the optical modes at Γ into TO (371 cm^{-1}) and LO (680 cm^{-1}).

The two supercell sizes give almost the same dispersions. The largest differences are on zone-boundary acoustic branches: 18 cm^{-1} for the transverse acoustic mode of Si at X (129 to 147 cm^{-1}), 18 cm^{-1} for MgO and 8 cm^{-1} for Al. At 300 K , the heat capacity changes by less than $0.05\text{ J K}^{-1}\text{ mol}^{-1}$ between the two sizes, so the smaller supercell is enough for thermodynamic properties, while the larger one is needed for zone-boundary frequencies. The **g2k1** runs give the same frequencies as **g1** to within 0.02 cm^{-1} .

Cost on one and two GPUs. Table 14 gives the VASP wall time of each step of the workflows. A complete workflow takes between 28 min (Si) and 104 min (MgO) on one GPU. The force calculations in the larger supercell dominate, except for MgO, where the Born charges and ε_∞ (LEPSILON on the 2-atom cell, 21^3 k -mesh) take 31 min, 30% of the workflow and more than the two displacements of the 54-atom supercell. The phonopy post-processing takes about 10 s per supercell and is negligible.

Two GPUs shorten the workflows by a factor of 1.37 to 1.60, a parallel efficiency of 69 to 80%, so they cost 25 to 46% more GPU-hours. The gain depends on the size of the cell, as in Section 4.6: none for the relaxations of the 1- and 2-atom primitive cells (0.96 to $1.15\times$), 1.24 to $1.44\times$ for the 54-atom supercells, and 1.40 to $1.72\times$ for the 64- to 128-atom supercells.

Both campaigns were submitted together; the two-GPU campaign was queued behind the one-GPU campaign so that the two never ran at the same time. After a wait for earlier jobs of this study, the one-GPU campaign (three materials, one after the other) ran for 3.0 h, and the two-GPU campaign for 2.1 h. For phonons of small cells, one GPU is therefore the economical choice; two

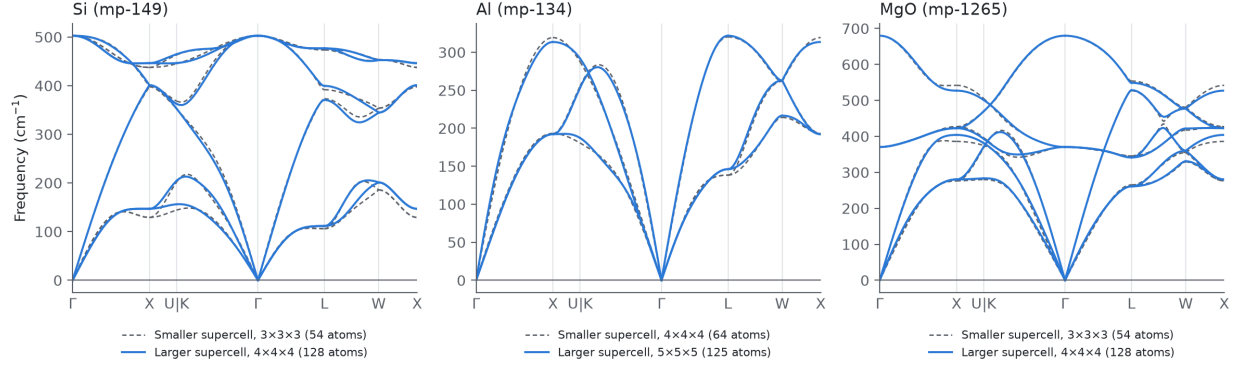


Figure 4: PBE phonon dispersions of Si, Al, and MgO (one GPU, **g1**), in cm^{-1} . Dashed gray: smaller supercell; solid blue: larger supercell (Table 6). The MgO bands include the non-analytical term correction. U|K marks a discontinuity of the path.

Table 14: VASP wall time of the phonon workflows (minutes) and GPU-hours, on one GPU (**g1**) and two GPUs (**g2k1**). Speedup: total **g1** time divided by total **g2k1** time.

Material	Variant	Relax.	Born	Small SC	Large SC	Total	GPU-h	Speedup
Si	g1	1.8	—	6.1	20.2	28.0	0.47	1.60
	g2k1	1.6	—	4.2	11.8	17.5	0.58	
Al	g1	2.9	—	14.2	31.4	48.5	0.81	1.49
	g2k1	3.1	—	10.1	19.4	32.5	1.08	
MgO	g1	7.6	31.5	20.0	44.9	103.8	1.73	1.37
	g2k1	7.0	23.0	16.1	29.5	75.6	2.52	

GPUs are worth using only to get the result about 1.5 times sooner when the node is otherwise idle.

4.9 Concurrent runs on one GPU

All 26 batches (100 calculations) completed, with no other job on the node. The energies of all copies are identical to those of the reference runs, and one copy under MPS runs as fast as without it (throughput 0.98 to 0.99 $\times$). Table 15 and Figure 5 give the throughput gains.

Table 15: Throughput gain $N t_{\text{ref}}/T_{\text{batch}}$ of N identical single points sharing one GPU, without and with MPS (ideal value: N). t_{ref} : time of one calculation alone.

Series	Material	Atoms	t_{ref} (s)	$N = 2$		$N = 4$		$N = 8$	
				no MPS	MPS	no MPS	MPS	no MPS	MPS
pbe_sp	Si	8	18.1	0.56	1.90	0.61	3.47	0.60	5.78
pbe_sp	MgO	32	138.1	0.55	1.89	0.58	3.31	0.56	5.04
r2scan_sp	Al	32	180.7	0.93	1.81	1.08	2.94	1.15	4.12
r2scan_sp	Si	64	324.2	1.11	1.67	1.35	2.53	–	–

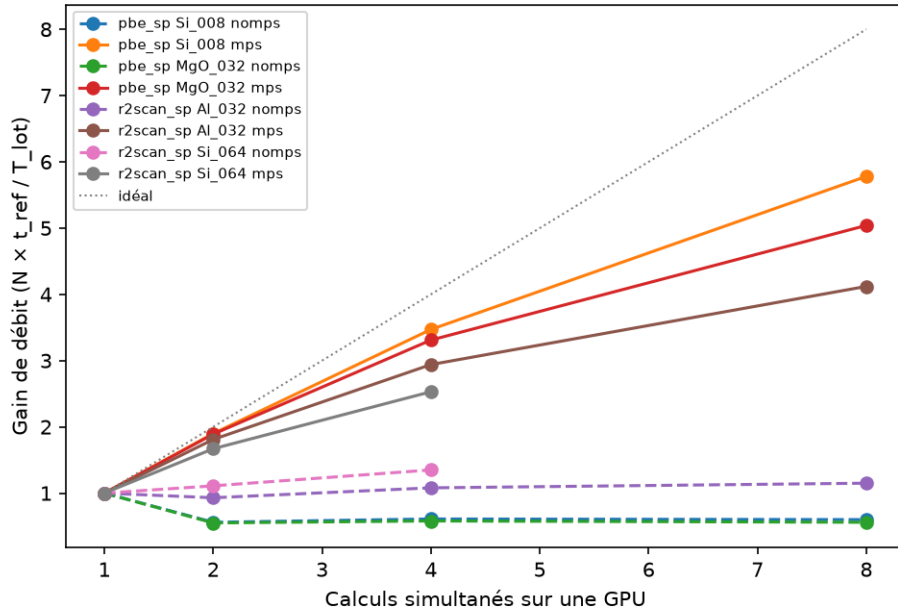


Figure 5: Throughput gain of N concurrent single points on one GPU. Solid lines: with MPS; dashed lines: without MPS; dotted line: ideal gain.

With MPS, throughput rises to 1.7–1.9 $\times$ with two calculations, 2.5–3.5 $\times$ with four, and 4.1–5.8 $\times$ with eight, a parallel efficiency of 52 to 72% at $N = 8$. Each calculation is slowed down by only 1.3 to 1.9 $\times$ at $N = 8$: the time per SCF step of PBE Si 8 atoms goes from 1.27 to 1.60s. The gain is largest for the smallest calculations, which leave most of the GPU idle, and decreases with system size. Mean GPU utilization rises from 25–38 % for one calculation to 69–90 % for eight, and

GPU memory grows in proportion to N , up to 34.5 GB for eight r²SCAN Al 32-atom copies, well within the 94 GB of the H100.

Without MPS, sharing the GPU brings nothing. For the PBE cases the batch is even 1.6 to 1.8 times slower than running the same calculations one after the other (gain 0.55 to 0.61), because each process waits for its time slice and the context switches add overhead: the time per SCF step of each copy grows about N -fold or more (PBE MgO: from 10.1 to 125 s at $N = 8$). For r²SCAN, whose kernels are longer, the gain stays between 0.93 and 1.35 $\times$.

5 Discussion and Recommendations

5.1 Are these GPUs suited to VASP?

The results do not show that the H100 GPUs are unsuited to VASP. They show that systems of 1 to 64 atoms leave these GPUs largely underused.

What the measurements show.

- **Absolute times are short.** A PBE single point on 64-atom MgO takes 163 s on one GPU, and a full r²SCAN relaxation of the same cell takes 48 min.
- **The GPUs are lightly loaded.** On one GPU, the mean utilization reported by `nvidia-smi` ranges from 10 to 50 %, and the peak GPU memory from 3 to 12 GB out of 94 GB. The utilization metric is the fraction of time during which at least one kernel is running, sampled every 2 s; it does not measure how much of the device each kernel occupies, so the actual occupancy is lower still.
- **Two GPUs bring a modest gain.** The speedup ranges from 1.0 to 1.9 $\times$, generally grows with system size, and vanishes for the 1-atom cell with `KPAR=1` (Section 4.6).

Interpretation. This is the expected behavior of plane-wave DFT on GPUs. For small cells, fast Fourier transforms and dense matrix operations are too small to fill an H100, and run time is dominated by latencies: kernel launches, synchronizations, host-device transfers, and, with two GPUs, inter-GPU communication. The GPU port becomes efficient for systems with a few hundred atoms, or with many bands and k -points.

Limits of the present study.

- **No CPU reference.** Only the GPU (OpenACC) build of VASP is installed on the node, so no CPU-only timings were measured. The present data therefore cannot support any claim about GPU versus CPU performance.
- **The CPU was not benchmarked.** Each calculation uses one MPI rank per GPU with `OMP_NUM_THREADS=1`: a single core does work out of the 16 allocated per GPU, which is the normal setup for the GPU build.
- **No large systems.** The largest cell has 64 atoms, below the size range where the GPU port is expected to pay off.

5.2 Recommendations for users of the node

- For cells of up to 64 atoms, one GPU per calculation is sufficient. Running two independent calculations, one per GPU, yields more throughput than one calculation on two GPUs.
- When two GPUs are used, prefer `KPAR=2` for cells with many irreducible k -points (here, up to about 32 atoms). With only 3 or 4 k -points (64-atom cells), `KPAR=1` can be as fast or faster, especially when the k -points do not split evenly between the two GPUs.
- Memory is not a constraint in this size range: the H100s leave ample room for much larger systems.
- For many small independent calculations (screening, phonon displacements), run 4 to 8 of them on the same GPU under MPS: throughput increases 2.5 to 5.8 times (Section 4.9). Without MPS, never run more than one calculation per GPU. On this node, MPS must be started inside the job, with pipe and log directories private to the job, as in the scripts of this study; the number of copies is limited by the host memory allocated per GPU.

5.3 Further tests

Two tests would complete the assessment of the hardware (the third one initially planned, concurrent runs under MPS, is reported in Section 4.9):

1. **CPU reference.** Build a CPU version of VASP 6.5.1 and rerun a subset of the calculations on 16 and 32 cores.
2. **Larger systems.** Extend the Si and MgO series to 128, 256, and 512 atoms, where the GPU port should be efficient.

6 Conclusion

On this dual-H100 node, the GPU build of VASP 6.5.1 runs the standard Materials Project PBE and r^2 SCAN workflows for 1- to 64-atom cells correctly and quickly: all 168 calculations completed, energies are consistent with MP once pseudopotential and k -point differences are accounted for, one-GPU and two-GPU energies are identical, and timings are reproducible to within 1 %. PBE single points and relaxations take seconds to a few minutes, and r^2 SCAN relaxations up to 48 min.

In this size range, however, one calculation uses only a small fraction of a GPU: utilization stays below about 50 %, memory below 13 GB, and the time per SCF step grows sublinearly with the number of atoms. Two GPUs shorten a calculation by a factor of 1.0 to 1.9 only, so for small cells running one calculation per GPU gives more throughput. When two GPUs are used, `KPAR=2` is preferable when there are many k -points, and `KPAR=1` can be better when there are only 3 or 4. Complete phonon workflows of small cells take less than two hours on one GPU. Most of the idle capacity can be recovered by running several calculations on the same GPU under MPS: eight concurrent single points give 4 to 6 times the throughput of one, while without MPS concurrent runs are no faster, or even slower, than running them one after the other. Larger systems remain to be tested.

Data and Code Availability

Scripts, inputs generation, and result tables are available at https://github.com/John-Akapulco/GPU_vmp_testing_VASP_etc. VASP output files are kept on the test machine; PAW potentials are not redistributed, in accordance with the VASP license.

A VASP input files (INCAR)

This appendix reproduces the INCAR files used in this work, as written by the generation scripts of the repository (pymatgen sorts the tags alphabetically). Within each series, all structures share the same template; the tags that change from one structure or variant to another are listed after each group of files.

A.1 Materials Project series: single points and relaxations

Files generated by `scripts/02_make_inputs.py`, shown here for MgO (8 atoms), one GPU (g1).

pbe_sp (MPStaticSet)

```
ALGO = Fast
EDIFF = 0.0004
ENCUT = 520.0
ISMear = -5
ISPIN = 2
KPAR = 1
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LORBIT = 11
LREAL = False
LVHAR = False
LVTOT = False
LWAVE = False
MAGMOM = 8*0.6
NELM = 100
NSW = 0
PREC = Accurate
SIGMA = 0.05
```

pbe_relax (MPRelaxSet)

```
ALGO = Fast
EDIFF = 0.0004
ENCUT = 520.0
IBRION = 2
ISIF = 3
ISMear = -5
ISPIN = 2
KPAR = 1
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LORBIT = 11
LREAL = Auto
LVHAR = False
LVTOT = False
LWAVE = False
MAGMOM = 8*0.6
NELM = 100
NSW = 99
PREC = Accurate
SIGMA = 0.05
```

r2scan_sp (MPScanStaticSet)

```
ALGO = Fast
EDIFF = 1e-05
ENCUT = 680.0
ISMear = -5
ISPIN = 2
KPAR = 1
KSPACING = 0.44
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LMIXTAU = True
LORBIT = 11
LREAL = False
LVHAR = False
LVTOT = False
LWAVE = False
MAGMOM = 8*0.6
METAGGA = R2scan
NELM = 200
NSW = 0
PREC = Accurate
SIGMA = 0.05
```

r2scan_relax (MPScanRelaxSet)

```
ALGO = All
EDIFF = 1e-05
EDIFFG = -0.02
ENCUT = 680.0
IBRION = 2
ISIF = 3
ISMear = -5
ISPIN = 2
KPAR = 1
KSPACING = 0.44
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LMIXTAU = True
LORBIT = 11
LREAL = Auto
LVHAR = False
LVTOT = False
LWAVE = False
MAGMOM = 8*0.6
METAGGA = R2scan
NELM = 200
NSW = 99
PREC = Accurate
SIGMA = 0.05
```

Tags that depend on the structure or the variant.

- KPAR: 1 for g1 and g2k1, 2 for g2k2 (Section 3.3).
- MAGMOM: $N \times 0.6$, with N the number of atoms (MP default initial moment).
- EDIFF (PBE only): $5 \times 10^{-5} \times N$ eV, i.e. from 5×10^{-5} (Al, 1 atom) to 3.2×10^{-3} eV (64 atoms). The r²SCAN sets use 1×10^{-5} eV for every cell.
- Smearing, PBE: ISMEAR=-5, SIGMA=0.05 everywhere, except Al 64 atoms (fewer than 4 irreducible k -points), where ISMEAR=0.
- Smearing and sampling, r²SCAN: KSPACING=0.44 (MgO), 0.287 (Si), and $0.22 \text{ \AA}^{-1}$ (Al, metal). For Al, SIGMA=0.2, with ISMEAR=2 (Methfessel-Paxton) in relaxations; the single points keep ISMEAR=-5.

The concurrent runs of Section 3.6 reuse, byte for byte, the INCAR files of the corresponding g1 single points.

A.2 Finite-displacement phonons

Files written by the phonon workflow (`scripts/05_phonon_setup.py`), shown for MgO. The relaxation file is the same for every pass (`relax1`, `relax2`, ...); the supercell file is the same for every displacement and both supercell sizes.

Tight relaxation

```
ADDGRID = True
ALGO = Normal
EDIFF = 1e-08
EDIFFG = -0.001
ENCUT = 520.0
IBRION = 2
ISIF = 3
ISMEAR = 0
ISPIN = 1
KPAR = 1
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LREAL = False
LVHAR = False
LVTOT = False
LWAVE = False
NELM = 200
NSW = 100
PREC = Accurate
SIGMA = 0.01
```

Born charges, ϵ_∞

```
ADDGRID = True
ALGO = Normal
EDIFF = 1e-08
ENCUT = 520.0
IBRION = -1
ISIF = 2
ISMEAR = 0
ISPIN = 1
KPAR = 1
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LEPSILON = True
LREAL = False
LVHAR = False
LVTOT = False
LWAVE = False
NELM = 200
NSW = 0
PREC = Accurate
SIGMA = 0.01
```

Displaced supercell

```
ADDGRID = True
ALGO = Normal
EDIFF = 1e-08
ENCUT = 520.0
IBRION = -1
ISIF = 2
ISMEAR = 0
ISPIN = 1
KPAR = 1
LAECHG = False
LASPH = True
LCHARG = False
LELF = False
LREAL = False
LVHAR = False
LVTOT = False
LWAVE = False
NELM = 200
NSW = 0
PREC = Accurate
SIGMA = 0.01
```

Differences between materials. Si uses the same files as MgO, without the Born step. For Al (metal), ISMEAR=1 and SIGMA=0.2 replace ISMEAR=0 and SIGMA=0.01 in the relaxation and supercell files. KPAR=1 in both the `g1` and `g2k1` variants.

A.3 Ab initio molecular dynamics

NVT molecular dynamics of LiN_3 (mp-2659, $2 \times 3 \times 3$ supercell, 144 atoms, Γ point only), prepared by `scripts/08_aimd_setup.py`. This campaign is in progress and is not discussed in the main text. The template below (`input/INCAR`) is the same for the four runs; the runs with a machine-learned force field (`calib_ml_omp1_g1`, `nvt_ml_g1`) add the two `ML_` tags.

Template, pure AIMD

```
ALGO = Fast
EDIFF = 1e-06
ENCUT = 520.0
IBRION = 0
ISIF = 2
ISMear = 0
ISPIN = 2
ISYM = 0
IVDW = 12
KPAR = 1
LANGEVIN_GAMMA = 10 10
LCHARG = False
LREAL = Auto
LWAVE = False
MDALGO = 3
NELM = 100
NELMIN = 4
POTIM = 1.0
PREC = Normal
RANDOM_SEED = 271828 0 0
SIGMA = 0.05
SYSTEM = LiN3 2x3x3 NVT
```

Additional tags, MLFF runs

```
ML_LMLFF = True
ML_MODE = train
```

Added to each segment (seg-XX/INCAR, by scripts/aimd_pipeline.py)

```
NSW = steps per segment
TEBEG = start temperature
TEEND = end temperature
```

Runs. `calib_dft_gam_g1` and `calib_dft_std_g1`: 200 steps at 300 K (NSW=200, TEBEG=TEEND=300), identical inputs run with `vasp_gam` and `vasp_std`. `calib_ml_omp1_g1` and `nvt_ml_g1`: 15 segments of 1000 steps (1ps) with a linear ramp from 300 to 500 K; each segment restarts from the previous `CONTCAR` and `ML_ABN`. The calibration run `calib_ml_omp1_g1` is limited to the first segment (300 to 313 K).

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